

KAVYA SREE MANIGA

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with 6+ years of hands-on experience designing and delivering scalable, high-quality data solutions and large-scale data pipelines that enable intelligent decision-making. Deep expertise in Python, PySpark, SQL, Apache Spark, Airflow, dbt, Snowflake, and AWS (S3, EMR, Glue), with strong proficiency in Hadoop ecosystem technologies and open table formats including Iceberg, Delta Lake, and Apache Parquet. Proven track record building Medallion Architecture data products, managing complex ETL/ELT workflows, and delivering well-structured, future-ready data solutions across on-premises and cloud environments. Actively leverages generative AI tools including Claude Code and GitHub Copilot to generate and refactor PySpark jobs, Hive and Snowflake SQL, Airflow DAG scaffolding, and documentation while retaining full ownership of review, hardening, validation, and production readiness. Experienced defining Claude agents and skills for recurring workflows including code review, PySpark job generation, backfill planning, and incident triage. Strong communicator who converts ambiguous business requirements into clear technical implementation plans and manages stakeholder communications with clarity and precision.

TECHNICAL SKILLS

Languages: Python (production-grade, 5+ years) | SQL (Advanced, CTEs, Window Functions, Stored Procedures, Query Optimization) | T-SQL | PySpark | Scala | Spark SQL | Bash | PowerShell

Hadoop Ecosystem and Big Data: Apache Spark | PySpark | Hadoop | Hive | HBase concepts | HDFS | MapReduce | Databricks | EMR | Apache Kafka | Google Pub/Sub

Open Table Formats: Apache Iceberg | Delta Lake | Apache Parquet | Apache Avro | Columnar Storage | Schema Evolution

AWS Data Services: S3 | EMR | Glue (Glue Catalog) | Lambda | Redshift | Kinesis | Firehose | Athena | Lake Formation | Step Functions | IAM | CloudWatch

Airflow and Orchestration: Apache Airflow (DAG development, custom operators, execution monitoring, failure triage, SLA adherence, backfill patterns) | Azure Data Factory

Modern Data Stack: dbt (models, macros, tests, documentation, CI/CD) | Snowflake | BigQuery | Redshift | Fivetran | Hightouch (Reverse ETL)

Data Modeling and Architecture: Medallion Architecture (Bronze/Silver/Gold) | Dimensional Modeling | Star Schemas | Lakehouse Architecture | Delta Lake | Iceberg | Data Products

Databases: PostgreSQL | MongoDB | SQL Server | DynamoDB | Azure Synapse | Hive Metastore

Generative AI Tools: Claude Code (PySpark generation, DAG scaffolding, incident triage, documentation) | GitHub Copilot | Claude Agents and Skills | MCP Connectors | LangGraph | RAG | Claude API

Data Quality and Observability: Automated Monitoring and Alerting | Anomaly Detection | dbt Tests | Data Lineage | Observability Signals | Runbooks | SLA Management | Microsoft Purview | Dataplex

Data Access and Security: IAM Roles and Trust Relationships | Lake Formation | Cross-Account Data Access | RBAC | Row-Level Security | Encryption | Key Management | HIPAA

CI/CD and DevOps: Terraform | Git | GitHub | GitLab | Azure DevOps | CI/CD Pipelines | Docker | Code Reviews | Agile | Jira | Confluence

Cloud Platforms: AWS (S3, Glue, EMR, Lambda, Redshift, Kinesis) | GCP (BigQuery, Dataflow, Dataproc, Pub/Sub, Vertex AI, Dataplex) | Azure (Synapse, Data Factory, Databricks, Fabric, Data Lake Gen2)

PROFESSIONAL EXPERIENCE

Data Engineer | **Digimarc** | OR, USA | Apr 2024 - Present

- Architected and executed end-to-end GCP data platform migration, designing scalable ETL/ELT pipelines and data products using BigQuery, Dataflow, Dataproc, and Cloud Storage with Medallion Architecture patterns across bronze, silver, and curated gold layers, reducing pipeline processing time by 35% through optimized partitioning and parallel execution.
- Engineered real-time Change Data Capture streaming pipelines using Pub/Sub and Dataflow to ingest data from PostgreSQL and MongoDB into BigQuery, enabling low-latency analytics and reducing data freshness lag from hours to minutes.
- Designed and optimized Apache Spark jobs on Dataproc using PySpark and Scala, enhancing ingestion and partitioning strategies for BigQuery datasets and reducing Spark job execution time by 30% through executor tuning and resource allocation improvements.

- Implemented dbt as the core transformation layer on BigQuery, building 20+ modular, version-controlled data models with reusable macros, automated quality tests, comprehensive documentation, and CI/CD pipelines that reduced transformation debugging time by 40% and standardized deployment practices.
- Developed and owned Apache Airflow DAGs from scratch with custom operators, dependency management, dbt run/test integration, and automated alerting, achieving 99%+ pipeline SLA adherence across all production workflows and building operational runbooks for failure triage and backfill patterns.
- Migrated datasets to Snowflake implementing RBAC, row-level security, and dbt CI/CD pipelines, and implemented data governance using Dataplex for lineage tracking, IAM-based access controls, and encryption across all GCP platforms to achieve full audit readiness.
- Actively leveraged Claude Code and GitHub Copilot to generate and refactor PySpark jobs, Snowflake SQL, Airflow DAG scaffolding, and documentation while retaining full ownership of review, hardening, validation, and production readiness, reducing new pipeline development time by 30 to 40%.
- Defined Claude agents and skills for recurring workflows including code review automation, PySpark job generation, backfill planning, and incident triage, integrating Claude Code with GitHub and Airflow through MCP connector patterns.
- Drove engineering best practices through GitHub workflows, code reviews, automated dbt tests, schema controls, and safe deployment patterns, partnering with BI developers to build Looker dashboards enabling self-serve reporting for 5+ teams and reducing ad-hoc requests by 25%.

Data Engineer | Wells Fargo | MD, USA | Nov 2022 - Dec 2023

- Architected and implemented Medallion Architecture (Bronze, Silver, Gold) across Azure Data Lakehouse using Microsoft Fabric and Azure Synapse Analytics, consolidating financial transaction data from 10+ internal and cloud sources into a unified enterprise data platform and improving data quality across 15+ regulated reporting pipelines.
- Built and optimized ETL/ELT pipelines using Azure Data Factory, PySpark, and Databricks to process high-volume structured and semi-structured financial data, implementing Python ETL jobs with automated monitoring and CI/CD automation via Azure DevOps, reducing manual data processing time by 50%.
- Engineered near-real-time financial data ingestion pipelines into Azure Data Lake Gen2 using event-driven architectures with Azure Event Hub and Data Factory blob triggers, processing high-velocity transaction data with fault-tolerant error handling and SLA-adherent alerting.
- Developed OLAP-based data marts, fact-dimension models, and star schemas in Azure Synapse supporting financial analytics and regulatory reporting, delivering Power BI dashboards integrated with Microsoft Fabric surfacing real-time revenue, spend, and risk insights.
- Modernized legacy SSIS packages to Azure Databricks using Terraform IaC and reusable CI/CD templates, reducing technical debt, and enforced financial regulatory compliance using RBAC, IAM controls, and Microsoft Purview for lineage tracking across all OneLake datasets.
- Built data quality checks, observability signals, and operational runbooks in Confluence covering ETL processes, pipeline standards, and on-call procedures, enabling independent troubleshooting and knowledge transfer across the data engineering team.

Data Engineer | HCLTech | India | Nov 2018 - Dec 2021

- Engineered end-to-end migration of clinical, claims, and administrative data from on-premise systems to AWS Redshift and S3, profiling data sources, performing source-to-target mapping, and designing scalable ETL pipelines using AWS Glue, PySpark on EMR, Lambda, and Step Functions that processed 200GB+ daily healthcare data with zero data loss.
- Integrated Epic EHR system data into Redshift and S3 using AWS Glue and event-driven Lambda triggers, enabling near-real-time reporting on patient visits, scheduling, care outcomes, and provider performance, reducing clinical reporting latency by over 40%.
- Developed Python-based ETL jobs on AWS EMR and Lambda processing high-volume structured and unstructured clinical and medical claims data including ICD and CPT healthcare reference data, implementing data quality routines for cleansing, confidence scoring, and anomaly detection using T-SQL and Python, reducing pipeline failures by 30%.
- Designed and built provider and member data models in AWS Redshift using dimensional modeling and star schemas, implementing MDM frameworks and source-to-target mappings integrating Oracle Cloud, SQL Server, and on-prem sources into unified analytical datasets supporting Cardiology Quality Assurance and Annual Outcomes reporting.
- Enforced HIPAA compliance and PHI data governance by configuring IAM roles, trust relationships, cross-account data access controls, encryption at rest and in transit, and key management across all AWS accounts, achieving 100% audit readiness.
- Built and evolved data quality checks, observability signals, and operational runbooks using AWS CloudWatch and Lambda-based alerting, proactively identifying data issues impacting care reporting and reducing mean time to resolution by 50%.

EDUCATION

Master of Professional Studies in Data Science | University of Maryland Baltimore County, Baltimore, USA

Bachelor of Technology in Civil Engineering | Manipal Institute of Technology, Karnataka, India